

INVERSE TRIGONOMETRIC FUNCTIONS 1+2+2+3(+3)

	0°	30°	45°	60°	90°	$15^\circ \left(\frac{\pi}{12}\right)$	$75^\circ \frac{5\pi}{12}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}-1}{2\sqrt{2}}$	$\frac{\sqrt{3}+1}{2\sqrt{2}}$
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$\frac{\sqrt{3}+1}{2\sqrt{2}}$	$\frac{\sqrt{3}-1}{2\sqrt{2}}$
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (or) $2-\sqrt{3}$	$\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (or) $2+\sqrt{3}$

$$\cos \pi = -1, \cos(\text{odd } \pi) = -1$$

$$\cos(\text{even } \pi) = +1$$

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

$$\cos 36^\circ = \frac{\sqrt{5}+1}{4}$$

$$\sin 54^\circ = \frac{\sqrt{5}+1}{4}$$

$$\cos 72^\circ = \frac{\sqrt{5}-1}{4}$$

$$\cos \frac{A}{2} = \sqrt{\frac{1+\cos A}{2}}$$

$$\sin \frac{A}{2} = \sqrt{\frac{1-\cos A}{2}}$$

$$\tan \frac{A}{2} = \sqrt{\frac{1-\cos A}{1+\cos A}}$$

$$\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\tan\theta}{1+\tan^2\theta}$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1 = \frac{1-\tan^2\theta}{1+\tan^2\theta}$$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

$1 - \cos 2\theta = 2\sin^2 \theta$; $1 - \cos \theta = 2\sin^2\left(\frac{\theta}{2}\right)$
 $1 + \cos 2\theta = 2\cos^2 \theta$; $1 + \cos \theta = 2\cos^2\left(\frac{\theta}{2}\right)$

$\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2\left(\frac{\theta}{2}\right)$

$\frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = \tan\left(\frac{\pi}{4} - \theta\right)$

$\left[\text{LHS} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = \frac{1 - \tan \theta}{1 + \tan \theta} = \frac{\tan \frac{\pi}{4} - \tan \theta}{1 + \tan \frac{\pi}{4} \tan \theta} \right]$

$\text{RHS} = \tan\left(\frac{\pi}{4} - \theta\right) = \text{RHS}$

$\frac{\cos 3\theta - \sin 3\theta}{\cos 3\theta + \sin 3\theta} = \tan\left(\frac{\pi}{4} - 3\theta\right)$

$\frac{\cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right)} = \tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)$

→ Trigonometric functions cannot have their inverses in the entire range of real numbers.
 → STF will be one-one funⁿ and onto funⁿ only in certain intervals and hence they have their inverses only in their corresponding domain and range.
 These values are called principal values.
 Eg: $y = \sin x \Rightarrow x = \sin^{-1} y$
 $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ - Range
 $y \in [-1, 1]$ - Domain

Table of (principal value branches) domain and range of ITRs

	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$\sec^{-1} x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$

Theorems:

- 1] (a) $\sin^{-1}(-x) = -\sin^{-1} x$; $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$
 (b) $\cos^{-1}(-x) = \pi - \cos^{-1} x$; $\sec^{-1}(-x) = \pi - \sec^{-1} x$
 (c) $\tan^{-1}(-x) = -\tan^{-1} x$; $\cot^{-1}(-x) = \pi - \cot^{-1} x$

- 2] (a) $\sin^{-1} x = \operatorname{cosec}^{-1}(\frac{1}{x})$; $\operatorname{cosec}^{-1} x = \sin^{-1}(\frac{1}{x})$
 (b) $\cos^{-1} x = \sec^{-1}(\frac{1}{x})$; $\sec^{-1} x = \cos^{-1}(\frac{1}{x})$
 (c) $\tan^{-1} x = \cot^{-1}(\frac{1}{x})$; $\cot^{-1} x = \tan^{-1}(\frac{1}{x})$

3] (a) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

(b) $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$

(c) $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$

4] (a) $\sin(\sin^{-1} x) = x$

(b) $\cos(\cos^{-1} x) = x$

(c) $\tan(\tan^{-1} x) = x$

5] (a) $\sin^{-1}(\sin \theta) = \theta$ iff $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

(b) $\cos^{-1}(\cos \theta) = \theta$ iff $0 \leq \theta \leq \pi$

(c) $\tan^{-1}(\tan \theta) = \theta$ iff $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$$6] (a) \tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \text{ if } xy < 1$$

$$(b) \tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) \text{ if } xy > 1$$

$$(c) \tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$$

$$(d) 2 \tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$(e) \tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\left(\frac{x+y+z-xyz}{1-(xy+yz+zx)}\right)$$

EXERCISE 2.1

$$\text{Eg 1]} \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

$$\begin{aligned} \text{** Eg 2]} \cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) &= \pi - \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) \\ &= \pi - \frac{\pi}{3} = \frac{2\pi}{3} \end{aligned}$$

$$(a) \cot^{-1}(-1) = \frac{3\pi}{4} = \left(\pi - \frac{\pi}{4}\right)$$

$$(b) \cot^{-1}(-\sqrt{3}) = \frac{5\pi}{6}$$

$$1] \sin^{-1}\left(-\frac{1}{2}\right) = -\sin^{-1}\frac{1}{2} = -\frac{\pi}{6}$$

$$2] \cos^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{6}$$

$$3] \operatorname{cosec}^{-1}(2) = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$4] \tan^{-1}(\sqrt{3}) = \tan^{-1}\sqrt{3} = \frac{\pi}{3}$$

$$5] \cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$6] \tan^{-1}(-1) = -\tan^{-1}1 = -\frac{\pi}{4}$$

$$7] \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

$$8] \cot^{-1}(\sqrt{3}) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$9] \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$10] \operatorname{cosec}^{-1}(-\sqrt{2}) = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\sin^{-1}\frac{1}{\sqrt{2}} = -\frac{\pi}{4}$$

Find the values of the following:

$$\begin{aligned} ** 11] \tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) \\ = \frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi + 2\pi}{4} = \frac{3\pi}{4} \end{aligned}$$

$$\begin{aligned} 12] \cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right) \\ = \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right) \\ = \frac{\pi}{2} + \frac{\pi}{6} = \frac{3\pi + \pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3} \end{aligned}$$

13] If $\sin^{-1}x = y$ then

$$(a) 0 \leq y \leq \pi$$

$$(c) 0 < y < \pi$$

$$(b) -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$(d) -\frac{\pi}{2} < y < \frac{\pi}{2}$$

$$\begin{aligned} ** 14] \tan^{-1}\sqrt{3} - \sec^{-1}(-2) \\ = \frac{\pi}{3} - [\pi - \sec^{-1}(2)] = \frac{\pi}{3} - \pi + \frac{\pi}{3} = \frac{\pi - 3\pi + \pi}{3} = -\frac{\pi}{3} \end{aligned}$$

EXERCISE 2.2

Prove the following:

Eg 3] (a) $\sin^{-1}(2x\sqrt{1-x^2}) = 2\sin^{-1}x$, $\left[-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}\right]$

$$\begin{aligned} \text{LHS} &= \sin^{-1}(2x\sqrt{1-x^2}), \text{ put } x = \sin\theta \\ &= \sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta}) \end{aligned}$$

$$\begin{aligned} &= \sin^{-1}(2\sin\theta\cos\theta) \\ &= \sin^{-1}(\sin 2\theta) \end{aligned}$$

$$= 2\theta = 2\sin^{-1}x = \text{RHS}$$

(b) $\sin^{-1}(2x\sqrt{1-x^2}) = 2\cos^{-1}x$, $\left[\frac{1}{\sqrt{2}} \leq x \leq 1\right]$

$$\begin{aligned} \text{LHS} &= \sin^{-1}(2x\sqrt{1-x^2}), \text{ put } x = \cos\theta \\ &= \sin^{-1}(2\cos\theta\sqrt{1-\cos^2\theta}) \quad \theta = \cos^{-1}x \\ &= \sin^{-1}(2\cos\theta\sin\theta) \\ &= \sin^{-1}(\sin 2\theta) \\ &= 2\theta = 2\cos^{-1}x = \text{RHS} \end{aligned}$$

Eg 7] $\tan^{-1}x + \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \tan^{-1}\left[\frac{3x-x^3}{1-3x^2}\right]$

$$\text{LHS} = \tan^{-1}x + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$= \tan^{-1}\left\{\frac{x + \frac{2x}{1-x^2}}{1 - \frac{2x^2}{1-x^2}}\right\}$$

$$= \tan^{-1}\left\{\frac{x-x^3+2x}{1-x^2-2x^2}\right\} = \tan^{-1}\left\{\frac{3x-x^3}{1-3x^2}\right\} = \text{RHS}$$

Eg 8] Find the value of $\cos(\sec^{-1}x + \operatorname{cosec}^{-1}x)$

$$= \cos \frac{\pi}{2} = 0$$

1] Prove:

$$3 \sin^{-1}x = \sin^{-1}(3x - 4x^3), \quad x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$\begin{aligned} \text{RHS} &= \sin^{-1}(3x - 4x^3), \text{ put } x = \sin \theta \\ &= \sin^{-1}(3 \sin \theta - 4 \sin^3 \theta) \quad \theta = \sin^{-1}x \\ &= \sin^{-1}(\sin 3\theta) \\ &= 3\theta = 3 \sin^{-1}x = \text{LHS} \end{aligned}$$

2] $3 \cos^{-1}x = \cos^{-1}(4x^3 - 3x), \quad x \in \left[\frac{1}{2}, 1\right]$

$$\begin{aligned} \text{RHS} &= \cos^{-1}(4x^3 - 3x), \text{ put } x = \cos \theta \\ &= \cos^{-1}(4 \cos^3 \theta - 3 \cos \theta) \quad \theta = \cos^{-1}x \\ &= \cos^{-1}(\cos 3\theta) \\ &= 3\theta = 3 \cos^{-1}x = \text{LHS} \end{aligned}$$

3] $\tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} = \tan^{-1} \frac{1}{2}$

$$\begin{aligned} \text{LHS} &= \tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} \quad \tan^{-1}x + \tan^{-1}y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \\ &= \tan^{-1} \left[\frac{\frac{2}{11} + \frac{7}{24}}{1 - \left(\frac{2}{11}\right)\left(\frac{7}{24}\right)} \right] = \tan^{-1} \left[\frac{\frac{48+77}{264}}{\frac{264-14}{264}} \right] \\ &= \tan^{-1} \left[\frac{48+77}{264-14} \right] = \tan^{-1} \left(\frac{125}{150} \right) = \tan^{-1} \frac{1}{2} = \text{RHS} \end{aligned}$$

Eg 4] $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{2}{11} = \tan^{-1} \frac{3}{4}$

$$\begin{aligned} \text{LHS} &= \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{2}{11} \\ &= \tan^{-1} \left[\frac{\frac{1}{2} + \frac{2}{11}}{1 - \frac{2}{22}} \right] = \tan^{-1} \left[\frac{\frac{11+4}{22}}{\frac{22-2}{22}} \right] = \tan^{-1} \left(\frac{15}{20} \right) \\ &= \tan^{-1} \left(\frac{3}{4} \right) = \text{RHS} \end{aligned}$$

$$*] \tan^{-1} 2 + \tan^{-1} 3 = \tan^{-1} \left(\frac{5}{1-6} \right) \neq \tan^{-1}(-1) \quad [2 \times 3 = 6 > 1]$$

$$= \pi + \tan^{-1} \left[\frac{2+3}{1-6} \right]$$

$$= \pi + \tan^{-1}(-1)$$

$$= \pi - \tan^{-1} 1$$

$$= \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Prove the following (or) Simplify

$$1] \sin^{-1}(2x\sqrt{1-x^2}) \text{ put } x = \sin \theta ; \theta = \sin^{-1} x$$

$$= \sin^{-1}(2 \sin \theta \sqrt{1-\sin^2 \theta})$$

$$= \sin^{-1}(2 \sin \theta \cos \theta)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$= 2\theta = 2 \sin^{-1} x$$

$$2] \sin^{-1}(2x\sqrt{1-x^2}) = 2 \cos^{-1} x \text{ put } x = \cos \theta ; \theta = \cos^{-1} x$$

$$= \sin^{-1}(2 \cos \theta \sqrt{1-\cos^2 \theta})$$

$$= \sin^{-1}(2 \cos \theta \sin \theta)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$= 2\theta = 2 \cos^{-1} x$$

$$3] \sin^{-1} \left(\frac{2x}{1+x^2} \right) \quad x = \tan \theta ; \theta = \tan^{-1} x$$

$$= \sin^{-1} \left(\frac{2 \tan \theta}{1+\tan^2 \theta} \right)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$= 2\theta = 2 \tan^{-1} x$$

$$\begin{aligned}
 4] \cos^{-1}(1-2x^2) \quad x &= \sin \theta ; \theta = \sin^{-1} x \\
 &= \cos^{-1}(1-2\sin^2 \theta) \\
 &= \cos^{-1}(\cos 2\theta) \\
 &= 2\theta = 2\sin^{-1} x
 \end{aligned}$$

$$\begin{aligned}
 5] \cos^{-1}\left(\frac{2x}{1+x^2}\right) \quad x &= \tan \theta ; \theta = \tan^{-1} x \\
 &= \cos^{-1}\left(\frac{2\tan \theta}{1+\tan^2 \theta}\right) \\
 &= \cos^{-1}(\sin 2\theta) \\
 &= \cos^{-1}\left(\cos \left(\frac{\pi}{2} - 2\theta\right)\right) \\
 &= \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2\tan^{-1} x
 \end{aligned}$$

$$\begin{aligned}
 6] \sin^{-1}(2x^2-1) \quad x &= \cos \theta ; \theta = \cos^{-1} x \\
 &= \sin^{-1}(2\cos^2 \theta - 1) \\
 &= \sin^{-1}(\cos 2\theta) \\
 &= \sin^{-1}\left(\sin \left(\frac{\pi}{2} - 2\theta\right)\right) \\
 &= \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2\cos^{-1} x
 \end{aligned}$$

$$\begin{aligned}
 7] \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \quad x &= \tan \theta ; \theta = \tan^{-1} x \\
 &= \cos^{-1}\left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta}\right) \\
 &= \cos^{-1}(\cos 2\theta) \\
 &= 2\theta = 2\tan^{-1} x
 \end{aligned}$$

$$8] \tan^{-1}\left(\frac{2x}{1-x^2}\right) \quad x = \tan \theta ; \theta = \tan^{-1} x$$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right) \\
 &= \tan^{-1} (\tan 2\theta) \\
 &= 2\theta = 2 \tan^{-1} x
 \end{aligned}$$

9] Prove: $\sin^{-1}(3x - 4x^3) = 3 \sin^{-1} x$

$$\begin{aligned}
 \text{LHS} &= \sin^{-1}(3x - 4x^3) \quad x = \sin \theta ; \theta = \sin^{-1} x \\
 &= \sin^{-1}(3 \sin \theta - 4 \sin^3 \theta) \\
 &= \sin^{-1}(\sin 3\theta) \\
 &= 3\theta = 3 \sin^{-1} x
 \end{aligned}$$

10] Simplify: $\cos^{-1}(4x^3 - 3x)$ $x = \cos \theta ; \theta = \cos^{-1} x$

$$\begin{aligned}
 &= \cos^{-1}(4 \cos^3 \theta - 3 \cos \theta) \\
 &= \cos^{-1}(\cos 3\theta) \\
 &= 3\theta = 3 \cos^{-1} x
 \end{aligned}$$

11] $\tan^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right)$ $x = \tan \theta ; \theta = \tan^{-1} x$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right) \\
 &= \tan^{-1}(\tan 3\theta) \\
 &= 3\theta = 3 \tan^{-1} x
 \end{aligned}$$

NOTE: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$

$$2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$$

s	b	h
3	4	5
8	15	17
13	84	85
5	12	13
56	33	65

EXERCISE 2.2

** 4) $2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{31}{17}$

$$\text{LHS} = 2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7}$$

$$= \tan^{-1} \left[\frac{2(\frac{1}{2})}{1 - (\frac{1}{2})^2} \right] + \tan^{-1} \frac{1}{7}$$

$$= \tan^{-1} \left(\frac{4}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right)$$

$$= \tan^{-1} \left[\frac{\frac{4}{3} + \frac{1}{7}}{1 - (\frac{4}{3})(\frac{1}{7})} \right]$$

$$= \tan^{-1} \left(\frac{28+3}{21-4} \right) = \tan^{-1} \left(\frac{31}{17} \right) = \text{RHS}$$

Mis Egro] Show that $\sin^{-1} \frac{3}{5} - \sin^{-1} \frac{8}{17} = \cos^{-1} \frac{84}{85}$

$$\text{LHS} = \sin^{-1} \frac{3}{5} - \sin^{-1} \frac{8}{17}$$

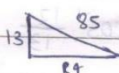


$$\sin^{-1} \frac{3}{5} = \tan^{-1} \frac{3}{4} \quad \sin^{-1} \frac{8}{17} = \tan^{-1} \frac{8}{15}$$

$$= \tan^{-1} \frac{3}{4} - \tan^{-1} \frac{8}{15}$$

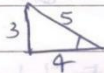
$$= \tan^{-1} \left(\frac{\frac{3}{4} - \frac{8}{15}}{1 + \frac{24}{60}} \right) = \tan^{-1} \left(\frac{45-32}{60+24} \right)$$

$$= \tan^{-1} \left(\frac{13}{84} \right)$$

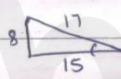
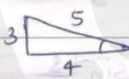


$$= \cos^{-1} \left(\frac{84}{85} \right)$$

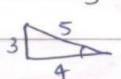
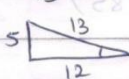
Mis Ex 3] PT : $2 \sin^{-1} \frac{3}{5} = \tan^{-1} \frac{24}{7}$

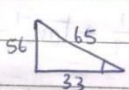
$$\begin{aligned} \text{LHS} &= 2 \sin^{-1} \frac{3}{5} \\ &= 2 \tan^{-1} \frac{3}{4} \\ &= \tan^{-1} \left(\frac{2 \left(\frac{3}{4} \right)}{1 - \frac{9}{16}} \right) \\ &= \tan^{-1} \left(\frac{\frac{3}{2} \cdot \frac{16}{7}}{1 - \frac{9}{16}} \right) \\ &= \tan^{-1} \left(\frac{24}{7} \right) = \text{RHS} \end{aligned}$$


Mis Ex 4] PT : $\sin^{-1} \frac{8}{17} + \sin^{-1} \frac{3}{5} = \tan^{-1} \frac{77}{36}$

$$\begin{aligned} \text{LHS} &= \sin^{-1} \frac{8}{17} + \sin^{-1} \frac{3}{5} \\ &= \tan^{-1} \frac{8}{15} + \tan^{-1} \frac{3}{4} \\ &= \tan^{-1} \left(\frac{\frac{8}{15} + \frac{3}{4}}{1 - \frac{24}{60}} \right) = \tan^{-1} \left(\frac{32+45}{60-24} \right) \\ &= \tan^{-1} \left(\frac{77}{36} \right) = \text{RHS} \end{aligned}$$



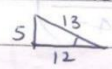
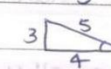
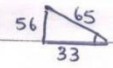
Mis Ex 5] PT : $\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} = \cos^{-1} \frac{33}{65}$

$$\begin{aligned} \text{LHS} &= \cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} \\ &= \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{5}{12} \\ &= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{15}{48}} \right) = \tan^{-1} \left(\frac{16+25}{48-15} \right) = \tan^{-1} \left(\frac{41}{33} \right) \end{aligned}$$



$$\begin{aligned}
 &= \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{5}{12} \\
 &= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{15}{48}} \right) = \tan^{-1} \left(\frac{36 + 20}{48 - 15} \right) \\
 &= \tan^{-1} \left(\frac{56}{33} \right)
 \end{aligned}$$


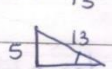
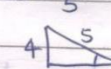
$$= \cos^{-1} \left(\frac{33}{65} \right) = \text{RHS}$$

MisEx 6] PT: $\cos^{-1} \frac{12}{13} + \sin^{-1} \frac{3}{5} = \sin^{-1} \frac{56}{65}$

$$\begin{aligned}
 \text{LHS} &= \cos^{-1} \frac{12}{13} + \sin^{-1} \frac{3}{5} \\
 &= \tan^{-1} \frac{5}{12} + \tan^{-1} \frac{3}{4} \\
 &= \tan^{-1} \left(\frac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{15}{48}} \right) \\
 &= \tan^{-1} \left(\frac{20 + 36}{48 - 15} \right) = \tan^{-1} \left(\frac{56}{33} \right)
 \end{aligned}$$




$$= \sin^{-1} \left(\frac{56}{65} \right) = \text{RHS}$$

MisEx 7] $\tan^{-1} \frac{63}{16} = \sin^{-1} \frac{5}{13} + \cos^{-1} \frac{3}{5}$

$$\begin{aligned}
 \text{RHS} &= \sin^{-1} \frac{5}{13} + \cos^{-1} \frac{3}{5} \\
 &= \tan^{-1} \frac{5}{12} + \tan^{-1} \frac{4}{3} \\
 &= \tan^{-1} \left(\frac{\frac{5}{12} + \frac{4}{3}}{1 - \frac{20}{36}} \right) = \tan^{-1} \left(\frac{15 + 48}{36 - 20} \right)
 \end{aligned}$$



~~$\tan^{-1}$~~

$$= \tan^{-1}\left(\frac{63}{16}\right) = \text{LHS}$$

Mis Ex 8] $\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8} = \frac{\pi}{4}$

$$\text{LHS} = \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8}$$

$$= \tan^{-1}\left(\frac{\frac{1}{5} + \frac{1}{7}}{1 - \frac{1}{35}}\right) + \tan^{-1}\left(\frac{\frac{1}{3} + \frac{1}{8}}{1 - \frac{1}{24}}\right)$$

$$= \tan^{-1}\left(\frac{7+5}{35-1}\right) + \tan^{-1}\left(\frac{8+3}{24-1}\right)$$

$$= \tan^{-1}\left(\frac{12}{34}\right) + \tan^{-1}\left(\frac{11}{23}\right)$$

$$= \tan^{-1}\left(\frac{6}{17}\right) + \tan^{-1}\left(\frac{11}{23}\right)$$

$$= \tan^{-1}\left(\frac{\frac{6}{17} + \frac{11}{23}}{1 - \frac{66}{391}}\right) = \tan^{-1}\left(\frac{138+187}{391-66}\right)$$

$$= \tan^{-1}\left(\frac{325}{325}\right) = \tan^{-1}(1) = \frac{\pi}{4} = \text{RHS}$$

Write the following funⁿ in the simplest form:

** 5] $\tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right]$ $x = \tan \theta$; $\theta = \tan^{-1} x$

$$= \tan^{-1}\left[\frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta}\right] = \tan^{-1}\left(\frac{\sec \theta - 1}{\tan \theta}\right)$$

$$= \tan^{-1}\left[\frac{\frac{1}{\cos \theta} - 1}{\frac{\sin \theta}{\cos \theta}}\right] = \tan^{-1}\left(\frac{1 - \cos \theta}{\sin \theta}\right)$$

$$= \tan^{-1} \left[\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right] = \tan^{-1} \left[\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right]$$

$$= \tan^{-1} \left[\tan \left(\frac{\theta}{2} \right) \right] = \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x$$

Mis Ex 11) $\tan^{-1} \left[\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right]$ put $x = \cos 2\theta$
 $\theta = \frac{1}{2} \cos^{-1} x$

$$= \tan^{-1} \left[\frac{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{2 \cos^2 \theta} - \sqrt{2 \sin^2 \theta}}{\sqrt{2 \cos^2 \theta} + \sqrt{2 \sin^2 \theta}} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{2} (\cos \theta - \sin \theta)}{\sqrt{2} (\cos \theta + \sin \theta)} \right] \quad \div \text{Num \& Deno by 'cos } \theta'$$

$$= \tan^{-1} \left[\frac{1 - \tan \theta}{1 + \tan \theta} \right]$$

$$= \tan^{-1} \left[\tan \left(\frac{\pi}{4} - \theta \right) \right]$$

$$= \frac{\pi}{4} - \theta = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x$$

*] $\tan^{-1} \left[\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right]$ put $x^2 = \cos 2\theta$
 $\theta = \frac{1}{2} \cos^{-1}(x^2)$

$$= \tan^{-1} \left[\frac{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{2 \cos^2 \theta} + \sqrt{2 \sin^2 \theta}}{\sqrt{2 \cos^2 \theta} - \sqrt{2 \sin^2 \theta}} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{2} (\cos \theta + \sin \theta)}{\sqrt{2} (\cos \theta - \sin \theta)} \right]$$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{1 + \tan \theta}{1 - \tan \theta} \right) \\
 &= \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \theta \right) \right] \\
 &= \frac{\pi}{4} + \theta = \frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2
 \end{aligned}$$

$$\star] \quad \tan^{-1} \left[\frac{\sqrt{1+x^3} - \sqrt{1-x^3}}{\sqrt{1+x^3} + \sqrt{1-x^3}} \right] \quad \text{put } x^3 = \cos 2\theta \\
 \theta = \frac{1}{2} \cos^{-1} x^3$$

$$= \tan^{-1} \left[\frac{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{2\cos^2 \theta} - \sqrt{2\sin^2 \theta}}{\sqrt{2\cos^2 \theta} + \sqrt{2\sin^2 \theta}} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{2}(\cos \theta - \sin \theta)}{\sqrt{2}(\cos \theta + \sin \theta)} \right]$$

$$= \tan^{-1} \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right)$$

$$= \tan^{-1} \left[\tan \left(\frac{\pi}{4} - \theta \right) \right]$$

$$= \frac{\pi}{4} - \theta = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x^3$$

$$\text{5] Simplify : } \tan^{-1} \left[\frac{\sqrt{1+x^2}-1}{x} \right] \quad \text{put } x = \tan \theta \\
 \theta = \tan^{-1} x$$

$$= \tan^{-1} \left[\frac{\sqrt{1+\tan^2 \theta} - 1}{\tan \theta} \right]$$

$$= \tan^{-1} \left[\frac{\sec \theta - 1}{\tan \theta} \right] = \tan^{-1} \left[\frac{\frac{1}{\cos \theta} - 1}{\frac{\sin \theta}{\cos \theta}} \right]$$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) = \tan^{-1} \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) \\
 &= \tan^{-1} \left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) \\
 &= \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x
 \end{aligned}$$

5] Simplify : $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ put $x = \tan \theta$
 $\theta = \tan^{-1} x$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta} \right) \\
 &= \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{\frac{1}{\cos \theta} - 1}{\frac{\sin \theta}{\cos \theta}} \right) \\
 &= \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) = \tan^{-1} \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) \\
 &= \tan^{-1} \left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) \\
 &= \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x
 \end{aligned}$$

Eg 5] Express $\tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right)$, $-\frac{3\pi}{2} < x < \frac{\pi}{2}$ in simplest form:

$$\begin{aligned}
 &\tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right) + \frac{\pi}{2} \\
 &= \tan^{-1} \left(\frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} - 2 \sin \frac{x}{2} \cos \frac{x}{2}} \right) \\
 &= \tan^{-1} \left(\frac{(\cos \frac{x}{2} - \sin \frac{x}{2})(\cos \frac{x}{2} + \sin \frac{x}{2})}{(\cos \frac{x}{2} - \sin \frac{x}{2})^2} \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \tan^{-1} \left[\frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right] \quad \div \text{Num \& Den} \\
 &\quad \quad \quad \text{by } \cos \frac{x}{2} \\
 &= \tan^{-1} \left[\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right] \\
 &= \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right] \\
 &= \frac{\pi}{4} + \frac{x}{2}
 \end{aligned}$$

Eg 5] Simplify : $\tan^{-1} \left[\frac{\cos x}{1 - \sin x} \right]$

$$\begin{aligned}
 &= \tan^{-1} \left[\frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} - 2 \sin \frac{x}{2} \cos \frac{x}{2}} \right] \\
 &= \tan^{-1} \left[\frac{(\cos \frac{x}{2} - \sin \frac{x}{2})(\cos \frac{x}{2} + \sin \frac{x}{2})}{(\cos \frac{x}{2} - \sin \frac{x}{2})^2} \right] \\
 &= \tan^{-1} \left[\frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right] \quad \div \text{Num \& Den} \\
 &\quad \quad \quad \text{by } \cos \frac{x}{2} \\
 &= \tan^{-1} \left[\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right] = \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right] \\
 &= \frac{\pi}{4} + \frac{x}{2}
 \end{aligned}$$

Eg 5(a)] Simplify $\tan^{-1} \left(\frac{\sin x}{1 - \cos x} \right)$

$$\begin{aligned}
 &= \tan^{-1} \left[\frac{x \sin \frac{x}{2} \cos \frac{x}{2}}{x \sin^2 \frac{x}{2}} \right] = \tan^{-1} \left[\cot \frac{x}{2} \right]
 \end{aligned}$$

$$= \tan^{-1} \left[\tan \left(\frac{\pi}{2} - \frac{x}{2} \right) \right] = \frac{\pi}{2} - \frac{x}{2}$$

Eg 5] Simplify $\tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right)$

$$= \tan^{-1} \left(\frac{\sin \left(\frac{\pi}{2} - x \right)}{1 - \cos \left(\frac{\pi}{2} - x \right)} \right)$$

$$= \tan^{-1} \left(\frac{\sin \left(\frac{\pi}{2} - x \right)}{2 \sin^2 \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

$$= \tan^{-1} \left(\frac{2 \sin \left(\frac{\pi}{4} - \frac{x}{2} \right) \cos \left(\frac{\pi}{4} - \frac{x}{2} \right)}{2 \sin^2 \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

$$= \tan^{-1} \left[\cot \left(\frac{\pi}{4} - \frac{x}{2} \right) \right] = \tan^{-1} \left[\tan \left(\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{x}{2} \right) \right) \right]$$

$$= \frac{\pi}{4} + \frac{x}{2}$$

6] $\tan^{-1} \frac{1}{\sqrt{x^2 - 1}}, |x| > 1$

put $x = \operatorname{cosec} \theta$; $\theta = \operatorname{cosec}^{-1} x$

$$= \tan^{-1} \left(\frac{1}{\sqrt{\operatorname{cosec}^2 \theta - 1}} \right) = \tan^{-1} \left(\frac{1}{\cot \theta} \right)$$

$$= \tan^{-1} (\tan \theta) = \theta = \operatorname{cosec}^{-1} x$$

** Eg 6] Write $\cot^{-1} \left(\frac{1}{\sqrt{x^2 - 1}} \right), x > 1$ in simplest form.

put $x = \sec \theta$; $\theta = \sec^{-1} x$

$$= \cot^{-1} \left(\frac{1}{\sqrt{\sec^2 \theta - 1}} \right) = \cot^{-1} \left(\frac{1}{\tan \theta} \right)$$

$$= \cot^{-1} (\cot \theta) = \theta = \sec^{-1} x$$

$$\begin{aligned}
 7] \quad \tan^{-1} \left(\sqrt{\frac{1-\cos x}{1+\cos x}} \right), 0 < x < \pi \\
 &= \tan^{-1} \left(\sqrt{\frac{2\sin^2 \frac{x}{2}}{2\cos^2 \frac{x}{2}}} \right) = \tan^{-1} \left(\sqrt{\tan^2 \frac{x}{2}} \right) \\
 &= \tan^{-1} \left(\tan \left(\frac{x}{2} \right) \right) = \frac{x}{2}
 \end{aligned}$$

$$\begin{aligned}
 *] \quad \tan^{-1} \left(\sqrt{\frac{1+\cos x}{1-\cos x}} \right) \\
 &= \tan^{-1} \left(\sqrt{\frac{2\cos^2 \frac{x}{2}}{2\sin^2 \frac{x}{2}}} \right) = \tan^{-1} \left(\sqrt{\cot^2 \frac{x}{2}} \right) \\
 &= \tan^{-1} \left(\cot \left(\frac{x}{2} \right) \right) = \tan^{-1} \left[\tan \left(\frac{\pi}{2} - \frac{x}{2} \right) \right] \\
 &= \frac{\pi}{2} - \frac{x}{2}
 \end{aligned}$$

$$\begin{aligned}
 9] \quad \tan^{-1} \left(\frac{x}{\sqrt{a^2-x^2}} \right), |x| < a \\
 \text{put } x = a \sin \theta, \theta = \sin^{-1} \left(\frac{x}{a} \right) \\
 &= \tan^{-1} \left(\frac{a \sin \theta}{\sqrt{a^2 - a^2 \sin^2 \theta}} \right) = \tan^{-1} \left(\frac{a \sin \theta}{a \cos \theta} \right) \\
 &= \tan^{-1} (\tan \theta) = \theta = \sin^{-1} \left(\frac{x}{a} \right)
 \end{aligned}$$

$$\begin{aligned}
 10] \quad \tan^{-1} \left(\frac{3a^2x-x^3}{a^3-3ax^2} \right), a > 0; -\frac{a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}} \\
 \text{put } x = a \tan \theta, \theta = \tan^{-1} \left(\frac{x}{a} \right)
 \end{aligned}$$

$$= \tan^{-1} \left(\frac{3a^3 \tan \theta - a^3 \tan^3 \theta}{a^3 - 3a^3 \tan^2 \theta} \right)$$

$\therefore$ Num & Den by a^3

$$= \tan^{-1} \left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right)$$

$$= \tan^{-1}(\tan 3\theta) = 3\theta = 3 \tan^{-1} \frac{x}{a}$$

Find the values of following:

$$11) \tan^{-1} \left[2 \cos \left(2 \sin^{-1} \frac{1}{2} \right) \right]$$

$$= \tan^{-1} \left[2 \cos \left(2 \cdot \frac{\pi}{6} \right) \right] = \tan^{-1} \left[2 \cos \left(\frac{\pi}{3} \right) \right]$$

$$= \tan^{-1} \left(2 \cdot \frac{1}{2} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$12) \cot \left(\tan^{-1} a + \cot^{-1} a \right)$$

$$= \cot \frac{\pi}{2} = 0$$

$$13) \tan^{-1} \frac{1}{2} \tan^{-1} \frac{1}{2} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], |x| < 1, y > 0, xy < 1$$

$$x = \tan A \quad \left| \quad y = \tan B \right.$$

$$A = \tan^{-1} x \quad \left| \quad B = \tan^{-1} y \right.$$

$$= \tan^{-1} \frac{1}{2} \left[\sin^{-1} \left(\frac{2 \tan A}{1 + \tan^2 A} \right) + \cos^{-1} \left(\frac{1 - \tan^2 B}{1 + \tan^2 B} \right) \right]$$

$$= \tan^{-1} \frac{1}{2} \left[\sin^{-1} (\sin 2A) + \cos^{-1} (\cos 2B) \right]$$

$$= \tan^{-1} \frac{1}{2} (2 \tan^{-1} x + 2 \tan^{-1} y) = \tan^{-1} \frac{1}{2} (2A + 2B)$$

$$= \tan(A+B)$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{x+y}{1-xy}$$

14] $\sin(\sin^{-1}\frac{1}{5} + \cos^{-1}x) = 1$, Find x .

$$= \sin^{-1}\frac{1}{5} + \cos^{-1}x = \sin^{-1}1$$

$$= \sin^{-1}\frac{1}{5} + \cos^{-1}x = \frac{\pi}{2}$$

By theorem, $\sin^{-1}\frac{1}{5} + \cos^{-1}\frac{1}{5} = \frac{\pi}{2}$

comparing, $x = \frac{1}{5}$

15] $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$, Find x .

$$\Rightarrow \tan^{-1}\left\{\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \frac{(x^2-1)}{x^2-4}}\right\} = \frac{\pi}{4}$$

$$\Rightarrow \frac{x^2 - x + 2x - 2 + x^2 + x - 2x - 2}{x^2 - 4 - x^2 + 1} = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{2x^2 - 4}{-3} = 1$$

$$\Rightarrow 2x^2 = -3 + 4$$

$$2x^2 = 1$$

$$x = \pm \frac{1}{\sqrt{2}}$$

Find the values of:

$$16] \sin^{-1}\left(\sin \frac{2\pi}{3}\right)$$

$$= \sin^{-1}(\sin 120^\circ)$$

$$= \sin^{-1}[\sin (180 - 60)]$$

$$= \sin^{-1}[\sin(60^\circ)] = 60^\circ = \frac{\pi}{3}$$

$$17] \tan^{-1}\left(\tan \frac{3\pi}{4}\right)$$

$$= \tan^{-1}(\tan 135^\circ)$$

$$= \tan^{-1}[\tan (180 - 45)]$$

$$= \tan^{-1}[-\tan(45)]$$

$$= \tan^{-1}(-1)$$

$$= -\frac{\pi}{4}$$

$$18] \tan \left[\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right]$$

$$= \tan (A+B)$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{6}{12}}$$

$$= \frac{9+8}{12-6} = \frac{17}{6}$$

$$\sin^{-1} \frac{3}{5} = A$$

$$\sin A = \frac{3}{5}$$

$$\tan A = \frac{3}{4}$$

$$\cot^{-1} \frac{3}{2} = B$$

$$\cot B = \frac{3}{2}$$

$$\tan B = \frac{2}{3}$$

$$*) \sin \left[\tan^{-1} \frac{3}{4} - \cos^{-1} \frac{5}{13} \right]$$

$$\begin{array}{l}
 \tan^{-1} \frac{3}{4} = A \quad \left| \quad \cos^{-1} \frac{5}{13} = B \right. \\
 \begin{array}{c} 3 \\ \hline 4 \end{array} \quad \frac{3}{4} = \tan A \quad \begin{array}{c} 12 \\ \hline 13 \end{array} \quad \frac{5}{13} = \cos B \\
 \sin A = \frac{3}{5} \quad \sin B = \frac{12}{13} \\
 \cos A = \frac{4}{5}
 \end{array}$$

$$\begin{aligned}
 &= \sin(A - B) \\
 &= \sin A \cos B - \cos A \sin B \\
 &= \left(\frac{3}{5}\right)\left(\frac{5}{13}\right) - \left(\frac{4}{5}\right)\left(\frac{12}{13}\right) \\
 &= \frac{15}{65} - \frac{48}{65} = -\frac{33}{65}
 \end{aligned}$$

$$\begin{aligned}
 *] \quad \cos\left(2 \cos^{-1} \frac{3}{5}\right) \\
 \cos^{-1} \frac{3}{5} = A \\
 \frac{3}{5} = \cos A \\
 &= 2 \cos^2 A - 1 \\
 &= 2\left(\frac{9}{25}\right) - 1 \\
 &= \frac{18 - 25}{25} = -\frac{7}{25}
 \end{aligned}$$

$$\begin{array}{l}
 *] \quad \cos\left(2 \tan^{-1} \frac{5}{12}\right) \quad \left| \quad \tan^{-1} \frac{5}{12} = A \right. \\
 \cos 2A \quad \frac{5}{12} = \tan A \\
 \frac{1 - \tan^2 A}{1 + \tan^2 A} \\
 = \frac{1 - \frac{25}{144}}{1 + \frac{25}{144}} = \frac{144 - 25}{144 + 25} = \frac{119}{169}
 \end{array}$$

$$*] \cos\left(\frac{1}{2} \cos^{-1} \frac{3}{5}\right)$$

$$= \cos\left(\frac{A}{2}\right) = \sqrt{\frac{1 + \cos A}{2}}$$

$$= \sqrt{\frac{1 + \frac{3}{5}}{2}} = \sqrt{\frac{8}{10}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$$

$$*] \sin\left[\frac{1}{2} \tan^{-1} \frac{4}{3}\right]$$

$$= \sin \frac{\theta}{2}$$

$$= \sqrt{\frac{1 - \cos \theta}{2}}$$

$$= \sqrt{\frac{1 - \frac{3}{5}}{2}} = \sqrt{\frac{2}{10}} = \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{5}}$$



$$\cos \theta = \frac{3}{5}$$

$$\text{Simplify: } \tan\left[\frac{1}{2} \left(\cos^{-1} \frac{12}{13}\right)\right]$$

$$= \tan\left(\frac{\theta}{2}\right)$$

$$= \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$= \sqrt{\frac{1 - \frac{12}{13}}{1 + \frac{12}{13}}} = \sqrt{\frac{13-12}{13+12}} = \sqrt{\frac{1}{25}} = \frac{1}{5}$$

$$\cos^{-1} \frac{12}{13} = \theta$$

$$\frac{12}{13} = \cos \theta$$

$$19] \text{ Find the value of } \cos^{-1}\left(\cos \frac{7\pi}{6}\right)$$

$$= \cos^{-1}(\cos 210^\circ)$$

$$= \cos^{-1}(\cos 180 + 30)$$

$$= \cos^{-1}(-\cos 30)$$

$$= \pi - \cos^{-1}(\cos 30)$$

$$= \pi - 30^\circ$$

$$= \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\begin{aligned}
 **20) \quad & \sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right) \\
 &= \sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right) \\
 &= \sin\left(\frac{\pi}{2}\right) = 1
 \end{aligned}$$

$$\begin{aligned}
 21) \quad & \tan^{-1}\sqrt{3} - \cot^{-1}(-\sqrt{3}) \\
 &= \tan^{-1}\sqrt{3} - (\pi - \cot^{-1}(\sqrt{3})) \\
 &= \tan^{-1}\sqrt{3} + \cot^{-1}\sqrt{3} - \pi \\
 &= \frac{\pi}{2} - \pi \\
 &= -\frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{Mis Eg 9)} \quad & \sin^{-1}\left(\sin \frac{3\pi}{5}\right) \\
 &= \sin^{-1}\left[\sin\left(\pi - \frac{2\pi}{5}\right)\right] \\
 &= \sin^{-1}\left[\sin\left(\frac{2\pi}{5}\right)\right] \quad (\text{less than } 90^\circ) \\
 &= \frac{2\pi}{5}
 \end{aligned}$$

$$\text{Mis Eg 12)} \quad \text{Simplify } \tan^{-1}\left[\frac{a \cos x - b \sin x}{b \cos x + a \sin x}\right], \text{ if } \frac{a}{b} \tan x > -1$$

÷ Num & Den by $b \cos x$

$$\begin{aligned}
 &= \tan^{-1}\left[\frac{\frac{a}{b} - \tan x}{1 + \frac{a}{b} \tan x}\right] \quad \left[\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)\right] \\
 &= \tan^{-1}\frac{a}{b} - \tan^{-1}(\tan x) \\
 &= \tan^{-1}\frac{a}{b} - x
 \end{aligned}$$

Mis Eg 13] Solve $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4} \rightarrow \textcircled{1}$

$$\Rightarrow \tan^{-1} \left(\frac{2x+3x}{1-6x^2} \right) = \frac{\pi}{4}$$

$$\Rightarrow \frac{2x+3x}{1-6x^2} = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{5x}{1-6x^2} = 1$$

$$\Rightarrow 5x = 1-6x^2$$

$$\Rightarrow 6x^2 + 5x - 1 = 0$$

$$\Rightarrow 6x^2 + 6x - x - 1 = 0$$

$$\Rightarrow 6x(x+1) - 1(x+1) = 0$$

$$\Rightarrow (6x-1)(x+1) = 0$$

$$\Rightarrow x = -1, \frac{1}{6}$$

$x = -1$ does not satisfy $\textcircled{1}$

$$\therefore \boxed{x = \frac{1}{6}}$$

Mis Ex 1] $\cos^{-1} \left[\cos \left(\frac{13\pi}{6} \right) \right]$

$$= \cos^{-1}(\cos 390^\circ)$$

$$= \cos^{-1}(\cos 360^\circ + 30^\circ)$$

$$= \cos^{-1}(\cos 30^\circ)$$

$$= 30^\circ = \frac{\pi}{6}$$

Mis Ex 2] $\tan^{-1} \left(\tan \frac{7\pi}{6} \right)$

$$= \tan^{-1}(\tan 210^\circ)$$

$$= \tan^{-1}(\tan (180^\circ + 30^\circ))$$

$$= \tan^{-1}(\tan 30^\circ) = 30^\circ = \frac{\pi}{6}$$

Mis Ex 9] $\tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right)$, $x \in [0, 1]$, Prove

TPF: $2 \tan^{-1} \sqrt{x} = \cos^{-1} \left(\frac{1-x}{1+x} \right)$

Put $x = \tan^2 \theta$; $\theta = \tan^{-1} \sqrt{x}$

$$\text{RHS} = \frac{1}{2} \cos^{-1} \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right)$$

$$= \frac{1}{2} \cos^{-1} (\cos 2\theta)$$

$$= \frac{1}{2} 2\theta = \theta = \tan^{-1} \sqrt{x} = \text{LHS}$$

Mis Ex 10] $\cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right] = \frac{x}{2}$, $x \in (0, \frac{\pi}{4})$

$$\left[\begin{aligned} \sqrt{1+\sin x} &= \sqrt{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}} \\ &= \sqrt{(\cos \frac{x}{2} + \sin \frac{x}{2})^2} = \cos \frac{x}{2} + \sin \frac{x}{2} \\ \text{Also } \sqrt{1-\sin x} &= \cos \frac{x}{2} - \sin \frac{x}{2} \end{aligned} \right]$$

$$\text{LHS} = \cot^{-1} \left[\frac{\cos \frac{x}{2} + \sin \frac{x}{2} + \cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2} - \cos \frac{x}{2} + \sin \frac{x}{2}} \right]$$

$$= \cot^{-1} \left[\frac{2 \cos \frac{x}{2}}{2 \sin \frac{x}{2}} \right] = \cot^{-1} \left[\cot \left(\frac{x}{2} \right) \right] = \frac{x}{2} = \text{RHS}$$

*] $\cot^{-1} \left[\frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\sqrt{1+\sin x} + \sqrt{1-\sin x}} \right]$

$$= \cot^{-1} \left[\frac{\cos \frac{x}{2} + \sin \frac{x}{2} - \cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2} + \cos \frac{x}{2} - \sin \frac{x}{2}} \right]$$

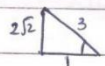
$$= \cot^{-1} \left(\frac{2 \sin \frac{x}{2}}{2 \cos \frac{x}{2}} \right) = \cot^{-1} \left(\tan \frac{x}{2} \right)$$

$$= \cot^{-1} \left(\cot \left(\frac{\pi}{2} - \frac{x}{2} \right) \right) = \frac{\pi}{2} - \frac{x}{2}$$

Mis Ex 12] $\frac{9\pi}{8} - \frac{9}{4} \sin^{-1} \frac{1}{3} = \frac{9}{4} \sin^{-1} \frac{2\sqrt{2}}{3}$; Prove.

Let it be $\frac{9\pi}{8} = \frac{9}{4} \sin^{-1} \frac{2\sqrt{2}}{3} + \frac{9}{4} \sin^{-1} \frac{1}{3}$

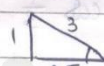
$$\text{RHS} = \frac{9}{4} \left[\sin^{-1} \frac{2\sqrt{2}}{3} + \sin^{-1} \frac{1}{3} \right]$$



$$\sin A = \frac{2\sqrt{2}}{3}$$

$$\tan A = \frac{2\sqrt{2}}{1}$$

$$A = \tan^{-1} 2\sqrt{2}$$



$$\sin A = \frac{1}{3}$$

$$\tan A = \frac{1}{2\sqrt{2}}$$

$$A = \tan^{-1} \frac{1}{2\sqrt{2}}$$

$$= \frac{9}{4} \left[\tan^{-1} (2\sqrt{2}) + \tan^{-1} \left(\frac{1}{2\sqrt{2}} \right) \right]$$

$$= \frac{9}{4} \tan^{-1} \left[\frac{2\sqrt{2} + \frac{1}{2\sqrt{2}}}{1 - 2\sqrt{2} \cdot \frac{1}{2\sqrt{2}}} \right]$$

$$= \frac{9}{4} \tan^{-1} \left[\frac{2\sqrt{2} + \frac{1}{2\sqrt{2}}}{0} \right]$$

$$= \frac{9}{4} \left[\tan^{-1} (\infty) \right]$$

$$= \frac{9}{4} \left(\frac{\pi}{2} \right)$$

$$= \frac{9\pi}{8} = \text{LHS}$$

(Q21)

$$\begin{aligned}
 \text{RHS} &= \frac{9}{4} \left[\sin^{-1} \frac{2\sqrt{2}}{3} + \sin^{-1} \frac{1}{3} \right] \\
 &= \frac{9}{4} \left[\sin^{-1} \frac{2\sqrt{2}}{3} + \cos^{-1} \frac{2\sqrt{2}}{3} \right] \quad \sin^{-1} \frac{1}{3} = \cos^{-1} \frac{2\sqrt{2}}{3} \\
 &= \frac{9}{4} \left(\frac{\pi}{2} \right) = \frac{9\pi}{8} = \text{LHS}
 \end{aligned}$$

(Q21)

$$\text{PT : } \frac{9\pi}{8} - \frac{9}{4} \sin^{-1} \frac{1}{3} = \frac{9}{4} \sin^{-1} \frac{2\sqrt{2}}{3}$$

$$\begin{aligned}
 \text{LHS} &= \frac{9}{4} \left[\frac{\pi}{2} - \sin^{-1} \frac{1}{3} \right] \\
 &= \frac{9}{4} \cos^{-1} \frac{1}{3} \\
 &= \frac{9}{4} \sin^{-1} \frac{2\sqrt{2}}{3} \\
 &= \text{RHS}
 \end{aligned}$$

Mis Ex 13] $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$; Solve : $x = ?$

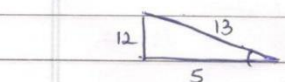
$$\Rightarrow \tan^{-1} \left(\frac{2 \cos x}{1 - \cos^2 x} \right) = \tan^{-1}(2 \operatorname{cosec} x)$$

$$\Rightarrow \frac{2 \cos x}{\sin^2 x} = \frac{2}{\sin x}$$

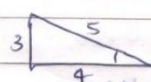
$$= \cot x = 1$$

$$x = \frac{\pi}{4}$$

Eg 11] Show that $\sin^{-1} \frac{12}{13} + \cos^{-1} \frac{4}{5} + \tan^{-1} \frac{63}{16} = \pi$



$$\sin^{-1} \frac{12}{13} = \tan^{-1} \frac{12}{5}$$



$$\cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4}$$

$$\begin{aligned}
 &\Rightarrow \tan^{-1} \frac{12}{5} + \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{63}{16} \\
 &= \tan^{-1} \left[\frac{\frac{12}{5} + \frac{3}{4}}{1 - \frac{36}{20}} \right] + \tan^{-1} \frac{63}{16} \\
 &= \tan^{-1} \left[\frac{48+15}{20-36} \right] + \tan^{-1} \frac{63}{16} \\
 &= \tan^{-1} \left(\frac{63}{-16} \right) + \tan^{-1} \frac{63}{16} \quad (xy > 1) \\
 &= \pi - \tan^{-1} \left(\frac{63}{16} \right) + \tan^{-1} \frac{63}{16} \\
 &= \pi = \text{RHS}
 \end{aligned}$$

Proofs:

(a) $\sin^{-1}(-x) = -\sin^{-1}x$

Let $\sin^{-1}(-x) = \theta \rightarrow (1)$

$-x = \sin \theta$

$x = -\sin \theta = \sin(-\theta)$

$\sin^{-1}x = -\theta$

$-\sin^{-1}x = \theta \rightarrow (2)$

$(1) \& (2) \Rightarrow \sin^{-1}(-x) = -\sin^{-1}x$

(c) $\tan^{-1}(-x) = -\tan^{-1}x$

Let $\tan^{-1}(-x) = \theta \rightarrow (1)$

$-x = \tan \theta$

$x = -\tan \theta = \tan(-\theta)$

$\tan^{-1}x = -\theta$

$-\tan^{-1}x = \theta \rightarrow (2)$

$(1) \& (2) \Rightarrow \tan^{-1}(-x) = -\tan^{-1}x$

$$b) \cos^{-1}(-x) = \pi - \cos^{-1}x$$

$$\text{let } \cos^{-1}(-x) = \theta \rightarrow (1)$$

$$-x = \cos \theta$$

$$x = -\cos \theta = \cos(\pi - \theta)$$

$$\cos^{-1}x = \pi - \theta$$

$$\theta = \pi - \cos^{-1}x \rightarrow (2)$$

$$(1) \& (2) \Rightarrow \cos^{-1}(-x) = \pi - \cos^{-1}x$$

$$d) \sin^{-1}x = \operatorname{cosec}^{-1}\frac{1}{x}$$

$$\text{let } \sin^{-1}x = \theta \rightarrow (1)$$

$$x = \sin \theta = \frac{1}{\operatorname{cosec} \theta}$$

$$\frac{1}{x} = \operatorname{cosec} \theta$$

$$\operatorname{cosec}^{-1}\frac{1}{x} = \theta \rightarrow (2)$$

$$(1) \& (2) \Rightarrow \sin^{-1}x = \operatorname{cosec}^{-1}\frac{1}{x}$$

$$e) \cos^{-1}x = \sec^{-1}\frac{1}{x}$$

$$\text{let } \cos^{-1}x = \theta \rightarrow (1)$$

$$x = \cos \theta = \frac{1}{\sec \theta}$$

$$\frac{1}{x} = \sec \theta$$

$$\sec^{-1}\frac{1}{x} = \theta \rightarrow (2)$$

$$(1) \& (2) \Rightarrow \cos^{-1}x = \sec^{-1}\frac{1}{x}$$

$$f) \tan^{-1}x = \cot^{-1}\frac{1}{x}$$

$$\text{let } \tan^{-1}x = \theta \rightarrow (1)$$

$$x = \tan \theta = \frac{1}{\cot \theta}$$

$$\frac{1}{x} = \cot \theta$$

$$\cot^{-1}\frac{1}{x} = \theta \rightarrow (2)$$

$$(1) \& (2) \Rightarrow \tan^{-1}x = \cot^{-1}\frac{1}{x}$$

$$g) \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$$

$$\text{let } \sin^{-1}x = \theta \rightarrow \textcircled{1}$$

$$x = \sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$$

$$\cos^{-1}x = \frac{\pi}{2} - \theta$$

$$\theta + \cos^{-1}x = \frac{\pi}{2}$$

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2} \text{ [using } \textcircled{1}]$$

$$h) \sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$$

$$\text{let } \sec^{-1}x = \theta \rightarrow \textcircled{1}$$

$$x = \sec \theta = \operatorname{cosec}\left(\frac{\pi}{2} - \theta\right)$$

$$\operatorname{cosec}^{-1}x = \frac{\pi}{2} - \theta$$

$$\theta + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$$

$$\sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$$

$$i) \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

$$\text{let } \tan^{-1}x = \theta \rightarrow \textcircled{1}$$

$$x = \tan \theta = \cot\left(\frac{\pi}{2} - \theta\right)$$

$$\cot^{-1}x = \frac{\pi}{2} - \theta$$

$$\theta + \cot^{-1}x = \frac{\pi}{2}$$

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

$$^{**}j) \tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$\text{Consider } \tan^{-1}x = A \quad \left| \quad \tan^{-1}y = B \right.$$

$$x = \tan A \quad \left| \quad y = \tan B \right.$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A+B) = \frac{x+y}{1-xy}$$

$$A+B = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$k) \tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$$

$$\text{Consider } \tan^{-1}x = A \quad \left| \quad \tan^{-1}y = B \right.$$

$$x = \tan A \quad \left| \quad y = \tan B \right.$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\tan(A-B) = \frac{x-y}{1+xy}$$

$$A-B = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$$

$$\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$$

$$d) \quad 2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$$

$$\text{consider, } \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$$

$$\text{put } y = x$$

$$\tan^{-1} x + \tan^{-1} x = \tan^{-1} \left[\frac{x+x}{1-x(x)} \right]$$

$$2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$$

$$m) \quad \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left[\frac{x+y+z-xyz}{1-(xy+yz+zx)} \right]$$

$$\text{LHS} = \tan^{-1} x + \tan^{-1} y + \tan^{-1} z$$

$$= \tan^{-1} \left(\frac{x+y}{1-xy} \right) + \tan^{-1} z$$

$$= \tan^{-1} \left\{ \frac{\frac{x+y}{1-xy} + z}{1 - \frac{xz+yz}{1-xy}} \right\}$$

$$= \tan^{-1} \left[\frac{x+y+z-xyz}{1-xy-yz-zx} \right] = \text{RHS}$$

$$n) \text{ (i) } 2 \tan^{-1} x = \sin^{-1} \left(\frac{2x}{1+x^2} \right) ; |x| \leq 1$$

$$\text{(ii) } 2 \tan^{-1} x = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) ; x \geq 0$$

$$\text{(iii) } 2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right) ; -1 < x < 1$$

$$(i) \text{ RHS} = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$\text{put, } x = \tan \theta$$

$$\theta = \tan^{-1} x$$

$$= \sin^{-1}\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$= 2\theta = 2 \tan^{-1} x = \text{LHS}$$

$$(ii) \text{ RHS} = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

$$\text{put, } x = \tan \theta ; \theta = \tan^{-1} x$$

$$= \cos^{-1}\left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}\right)$$

$$= \cos^{-1}(\cos 2\theta)$$

$$= 2\theta = 2 \tan^{-1} x = \text{LHS}$$

Mis Ex (14) Solve for x:

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2} \tan^{-1} x$$

$$2 \tan^{-1}\left(\frac{1-x}{1+x}\right) = \tan^{-1} x$$

$$\tan^{-1}\left[\frac{\frac{2-2x}{1+x}}{1 - \left(\frac{1-x}{1+x}\right)^2}\right] = \tan^{-1} x$$

$$\frac{2(1-x)}{(1+x)^2 - (1-x)^2} = x$$

$$2 - 2x = \frac{(1+x)^2}{4x}$$

$$(2-2x)(1+x) = 4x^2$$

$$\begin{aligned}
 2 + 2x - 2x - 2x^2 &= 4x^2 \\
 2 - 2x^2 &= 4x^2 \\
 \frac{2(1-x^2)}{4x} &= x \\
 2 - 2x^2 &= 4x^2 \leftarrow \\
 x &= \sqrt[3]{6}x^2 \\
 x^2 &= \frac{1}{3} \\
 x &= \pm \frac{1}{\sqrt{3}} \\
 \text{(or)} &= \frac{1-x^2}{2x} = \frac{x^2}{1-x^2} \\
 \text{put } x &= \tan \theta, \theta = \tan^{-1} x \\
 \tan^{-1} \left[\frac{\tan \frac{\pi}{4} - \tan \theta}{1 + \tan \frac{\pi}{4} \tan \theta} \right] &= \frac{1}{2} \theta \\
 \tan^{-1} \left[\tan \left(\frac{\pi}{4} - \theta \right) \right] &= \frac{1}{2} \theta \\
 \frac{\pi}{4} &= \frac{\theta}{2} + \theta \\
 \frac{\pi}{4} &= \frac{3\theta}{2} \\
 \frac{\pi}{6} &= \tan^{-1} x \\
 x &= \tan \frac{\pi}{6} \\
 x &= \frac{1}{\sqrt{3}}
 \end{aligned}$$

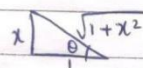
Mis Ex 15] $\sin(\tan^{-1} x), |x| < 1 =$

A) $\frac{x}{\sqrt{1-x^2}}$ B) $\frac{1}{\sqrt{1-x^2}}$ C) $\frac{1}{\sqrt{1+x^2}}$ D) $\frac{x}{\sqrt{1+x^2}}$

$\tan^{-1} x = \theta$

$\Rightarrow \frac{x}{1} = \tan \theta$

$\Rightarrow \sin \theta = \frac{x}{\sqrt{1+x^2}}$



Mis Ex 16] $\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$; $x = ?$

A) $0, \frac{1}{2}$ B) $1, \frac{1}{2}$ C) $0, \frac{1}{2}$ D) $\frac{1}{2}$

only $x=0$ satisfies the equation.

Mis Ex 17] $\tan^{-1} \frac{x}{y} - \tan^{-1} \frac{x-y}{x+y} =$

A) $\frac{\pi}{2}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{4}$ D) $-\frac{3\pi}{4}$

$$\Rightarrow \tan^{-1} \frac{x}{y} - \tan^{-1} \left[\frac{\frac{x}{x} - \frac{y}{x}}{\frac{x}{x} + \frac{y}{x}} \right]$$

$$\Rightarrow \tan^{-1} \frac{x}{y} - \tan^{-1} \left[\frac{1 - \frac{y}{x}}{1 + \frac{y}{x}} \right]$$

$$\Rightarrow \tan^{-1} \frac{x}{y} - \left[\tan^{-1} 1 - \tan^{-1} \frac{y}{x} \right]$$

$$\Rightarrow \tan^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x} - \tan^{-1} 1$$

$$\Rightarrow \tan^{-1} \frac{x}{y} + \cot^{-1} \frac{x}{y} - \tan^{-1} 1$$

$$\Rightarrow \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$